

Micro Entrepreneur Ledger Application

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Abstract—Abstract— Micro-entrepreneurs and small business owners often rely on notebooks or informal methods to manage daily financial records, making it difficult to track income, expenses, purchases, customer credit, and payments. This paper presents the Micro Entrepreneur Ledger App, a digital ledger system developed to simplify and organize financial record management for small-scale businesses. The proposed system allows users to create business profiles, record and categorize financial transactions, manage customer and supplier information, track credit and repayments, and view transaction history through a centralized dashboard. Each transaction is validated and stored in a relational database, while the stored data is processed to provide summarized financial information and business performance insights. The system follows a layered architecture consisting of a user interface, application and business logic layer, and database layer, implemented using Next.js, React, Prisma ORM, and SQLite. By replacing manual bookkeeping with a structured digital workflow, the proposed system improves accessibility, organization, and visibility of financial records for micro-entrepreneurs.

Index Terms—Micro Entrepreneurship, Digital Ledger, Financial Record Management, Small Business, Accounting Information System, Customer Credit, Transaction Management, Digital Financial Management

I. INTRODUCTION

Many micro-businesses still record daily sales, expenses, purchases, and customer credit using notebooks or basic spreadsheets. As transaction volume increases, these methods can make it difficult to maintain consistent records and obtain a clear view of the business's financial position.

Manual bookkeeping can introduce several difficulties. Transactions may be recorded inconsistently, calculations may contain errors, and older records may be difficult to retrieve. Tracking customer credit and outstanding payments can also become complicated when records are distributed across multiple notebooks or pages. These limitations can affect the ability of a small business owner to understand cash flow, expenses, and outstanding dues.

Digital financial management systems offer an organized alternative by storing business transactions electronically and making them easier to access when required. Studies on digital accounting adoption indicate that factors such as business readiness, usefulness, technology suitability, and data security can influence the acceptance of these systems among small businesses. Digital accounting tools can help small businesses organize their financial information systematically and make previously recorded business data easier to access and review [4].

The proposed Micro Entrepreneur Ledger App is designed specifically around the basic requirements of micro-entrepreneurs. The system enables users to create a business profile and record sales, income, expenses, purchases, customer credit, and payments. The stored transactions are organized into a digital ledger and summarized through a financial dashboard.

The primary objective of the system is to provide a simple and accessible bookkeeping solution without requiring advanced accounting knowledge. The application also provides a foundation for future integration with digital payment systems, cloud storage, automated reporting, and intelligent financial analysis.

II. LITERATURE SURVEY

A. Background

Digital financial systems have changed the way small businesses maintain and access their financial information. Compared with manual records, digital systems can organize transactions, simplify calculations, and make stored information easier to retrieve. Earlier research on digital accounting has considered how small businesses adopt these systems, with attention given to ease of use, business readiness, technological requirements, and the benefits perceived by users.

Research has also explored cloud-based accounting and digital payment systems for small and medium-sized businesses. These approaches show the potential for improving accessibility and reducing repeated manual work. Based on the reviewed studies, a useful micro-business ledger should focus on simple transaction recording, organized storage, financial summaries, and convenient access to business information.

B. Existing Approaches

Existing bookkeeping methods include notebooks, spreadsheets, dedicated accounting applications, and web-based systems. Manual records are simple to maintain but provide limited support for searching, calculations, and financial summaries. Spreadsheets provide better organization but still depend heavily on manual entry and formula management.

Dedicated accounting applications generally offer broader functionality, although some of their features may not be relevant to the simpler requirements of micro-businesses. Web-based systems provide greater accessibility and centralized data management. These observations led to the development of the proposed application, with emphasis on the essential financial activities of small-scale businesses.

III. MICRO ENTREPRENEUR LEDGER SYSTEM

The proposed Micro Entrepreneur Ledger App is a digital financial record management system designed for micro-entrepreneurs and small business owners. The application converts the traditional ledger workflow into a structured digital process.

The system consists of several interconnected stages:

- 1) Business registration and profile creation.
- 2) Recording sales and income.
- 3) Recording expenses and purchases.
- 4) Managing customer credit.
- 5) Recording customer payments.
- 6) Maintaining transaction history.
- 7) Calculating financial balances.
- 8) Displaying summarized information through a dashboard.

These stages provide an end-to-end workflow for maintaining basic business financial records.

A. Business and Transaction Input

The first stage of the application is business profile creation. The user enters basic information such as business name and other necessary profile details. The profile acts as the primary context for the transactions maintained by the application.

After creating a profile, users can enter different transaction categories. The major categories include sales, income, expenses, purchases, customer credit, and payments.

Each transaction can contain information such as transaction type, amount, date, description, and associated customer information where required. The structured input format allows transactions to be processed consistently.

Algorithm 1 Transaction Recording and Validation

```
1: function RECORD_TRANSACTION(transaction)
2:   validate transaction fields
3:   if transaction is invalid then
4:     display validation message
5:   return failure
6:   end if
7:   category ← classify(transaction)
8:   store(transaction, category)
9:   update_financial_summary(transaction)
10:  return success
11: end function
```

B. Transaction Processing and Validation

Transaction processing ensures that financial records are stored in a consistent format. Before storing a transaction, the application checks whether required fields are available and whether the entered amount is valid.

Transactions are categorized according to their purpose. Sales and income contribute to business inflow, while expenses and purchases represent financial outflow. Credit transactions are associated with customers and are maintained separately so that outstanding balances can be monitored.

Validation reduces the possibility of incomplete records and helps maintain consistency between transaction data and financial summaries.

Algorithm 2 Customer Credit Processing

```
1: function PROCESS_CREDIT(customer, amount)
2:   current_balance ← get_customer_balance(customer)
3:   new_balance ← current_balance + amount
4:   update_customer_balance(customer, new_balance)
5:   store_credit_transaction(customer, amount)
6:   return new_balance
7: end function
```

C. Ledger and Customer Credit Management

Customer credit management is an important component of the proposed application. Small businesses frequently allow customers to purchase goods or services on credit and collect payment later. Maintaining these records manually can make it difficult to determine outstanding amounts.

The proposed system associates credit transactions with customer records. When credit is added, the customer's outstanding balance is increased. When a payment is recorded, the outstanding balance is reduced.

This process provides a structured method for tracking customer dues and payment history.

Algorithm 3 Customer Payment Processing

```
1: function RECORD_PAYMENT(customer, payment)
2:   balance ← get_customer_balance(customer)
3:   updated_balance ← balance - payment
4:   if updated_balance ≤ 0 then
5:     updated_balance ← 0
6:   update_customer_balance(customer,
7:     updated_balance)
8:   store_payment(customer, payment)
9:   return updated_balance
10: end function
```

D. Complete System Workflow

The complete workflow begins when a user creates a business profile. The user then records business transactions through the appropriate input forms. The application validates each record and categorizes it according to the transaction type.

The validated transaction is stored in the corresponding data structure. Customer credit and payment transactions additionally update the customer's outstanding balance. The dashboard then aggregates the stored records to calculate sales, expenses, purchases, credit, payments, and net balance.

This architecture separates transaction entry from financial summarization and provides a foundation for future analytics.

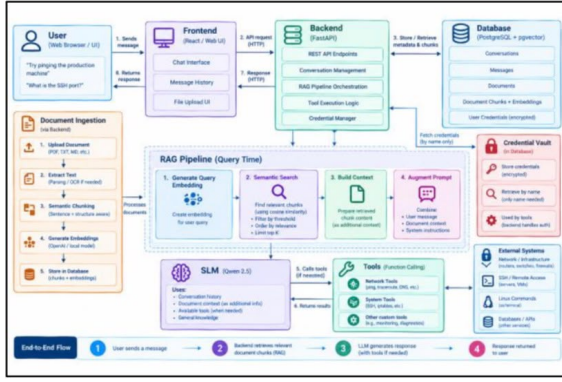


Fig. 1. Micro Entrepreneur Ledger App dashboard

IV. WORK DONE AND RESULTS ANALYSIS

The Micro Entrepreneur Ledger App was implemented to provide a structured digital method for managing the financial records normally maintained through traditional bookkeeping practices. The system focuses on the basic financial activities commonly required by micro-entrepreneurs, such as recording income, expenses, purchases, customer credit, and payments.

The application allows users to enter and manage their transactions in an organized manner. Each entry is checked and placed under the appropriate category before being stored. Customer credit and payment records are also updated whenever a transaction is made. The dashboard brings these records together and displays useful financial summaries, helping users get a clearer view of their business activities.

A. System Implementation and User Interface

The application is designed with a simple interface so that users can perform financial operations without extensive accounting knowledge. The main interface contains navigation options for transactions, customers, expenses, purchases, and dashboard information.

The transaction interface allows users to record business activities. The customer section provides access to customer information and credit balances. The dashboard presents summarized financial information.

The following figures represent the major application screens. If actual screenshots are available, they can be placed in the corresponding files inside the Images folder.

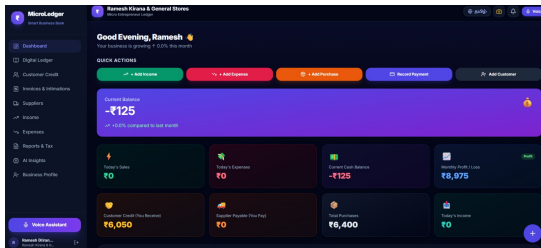


Fig. 2. Micro Entrepreneur Ledger App dashboard

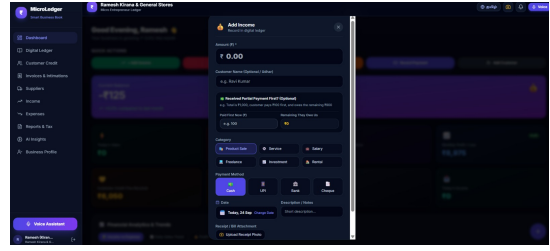


Fig. 3. Transaction recording interface

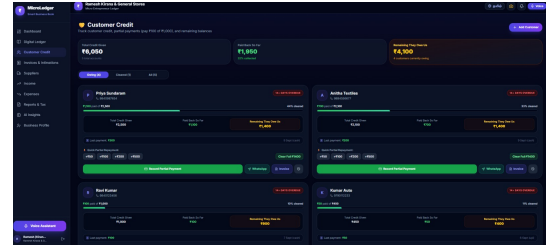


Fig. 4. Customer credit management interface

B. Functional and Performance Analysis

The application can be evaluated according to the correctness of transaction recording, balance calculation, customer credit management, payment tracking, and dashboard generation.

Functional testing verifies whether the application performs the required operations correctly. For example, a new sales transaction should be stored and included in the sales total. Similarly, an expense should increase the recorded expense value and affect the calculated balance.

Customer credit functionality can be evaluated by entering a credit transaction and subsequently recording a payment. The outstanding balance should change according to the recorded transactions.

The transaction history can be evaluated by entering multiple transactions and checking whether the records can be retrieved in the expected order.

The dashboard can be evaluated by comparing its displayed totals with the corresponding values calculated from the transaction records.

C. Use Cases and Societal Impact

The Micro Entrepreneur Ledger App can support different categories of small-scale business users.

Small Shop Owners: Shop owners can record daily sales, purchases, expenses, and customer credit in a single digital ledger.

Street Vendors: Vendors can maintain simple records of income and expenses without depending on physical notebooks.

Freelancers: Freelancers can use the application to record income, business expenses, and payment information.

Service Providers: Small service providers can track customer payments and outstanding amounts.

The application can also contribute to digital financial inclusion by making basic financial record management more accessible to users who may not have access to complex accounting systems.

D. Growth and Market Potential

The increasing use of digital financial technologies creates opportunities for applications that simplify financial management for small businesses. Digital accounting systems can provide advantages such as centralized records, easier retrieval, structured transaction management, and improved access to financial information.

The proposed application can be extended from a basic ledger into a broader financial management platform. Potential users include small shops, home-based businesses, vendors, freelancers, service providers, and other micro-enterprises.

The growth of the system can be evaluated using indicators such as number of registered businesses, number of transactions recorded, active users, payment records, report generation frequency, and user satisfaction.

Future versions can introduce cloud synchronization, digital payment integration, automated reporting, multilingual interfaces, and financial analytics.

E. System Modules and Data Management

The Micro Entrepreneur Ledger App is organized into several functional modules that support the complete bookkeeping workflow of a small business.

The User and Business Management Module stores basic business profile information and provides access to the ledger application. A business profile provides the context for the transactions maintained by the system.

The Transaction Management Module records sales, income, expenses, purchases, customer credit, and payments. Each transaction is stored with relevant information so that it can be retrieved and analyzed later.

The Customer Credit Module maintains information about credit transactions and payments received from customers. Whenever a new credit transaction is recorded, the corresponding customer balance is updated. Similarly, when a payment is entered, the outstanding amount is recalculated.

The Financial Dashboard Module aggregates transaction information and presents summarized business data. Important indicators include total sales, total expenses, purchases, outstanding credit, payments, and overall balance.

The Transaction History Module allows users to review previously entered records. This provides a convenient way to search and verify historical financial information.

The data layer can be organized into user and business profile records, transaction records, customer credit records, and analytics or report records. Separating these categories allows the application to manage different types of financial information systematically.

This structure also provides a foundation for future extensions such as cloud synchronization, automated reporting, and advanced financial analysis.

F. Testing and Validation

The application was evaluated using representative sales, income, expense, purchase, credit, and payment transactions. Input validation was tested by entering both complete and incomplete records. The system was also checked to ensure that transactions were assigned to the correct categories and stored correctly for later retrieval.

Balance calculations, customer credit updates, transaction history, and dashboard totals were compared with manually calculated reference values. The tests confirmed whether the application correctly processed the entered information and maintained consistency between stored transactions and the financial summaries presented to the user.

G. Security and Data Reliability

Business transaction records contain sensitive financial information, making appropriate protection and controlled access important for reliable system operation. The application should restrict access to business records through authenticated user accounts and ensure that users can access only the information associated with their business profile.

Transaction records should be validated before being stored to reduce the possibility of incorrect or incomplete financial data.

Data reliability can be improved through structured storage, consistent transaction formats, unique transaction identifiers, and regular backup mechanisms.

Cloud accounting research has identified security, organizational readiness, technology factors, and user expectations as important considerations when adopting digital accounting systems [1], [5].

The application can also maintain transaction timestamps and structured record identifiers to improve traceability. Such information can help users identify when transactions were created or updated and can support future auditing and reporting requirements.

If cloud-based storage is introduced, additional mechanisms such as secure authentication, encrypted communication, access control, and backup policies can be incorporated.

H. Limitations

Although the proposed Micro Entrepreneur Ledger App addresses several limitations of manual bookkeeping, the current system has some constraints.

The application focuses primarily on basic financial record management and does not provide complete enterprise-level accounting functionality. Automated tax filing, government platform integration, GST reporting, and advanced invoice management are outside the present scope.

The current evaluation also depends on test transactions rather than a large real-world dataset collected from multiple businesses. Therefore, measured results obtained during development may not represent long-term usage across different business environments.

Future evaluation using real transaction patterns from different categories of micro-businesses can provide a broader assessment.

The application may also require additional usability testing to identify differences in user requirements among shop owners, vendors, freelancers, and service providers.

Research on digital financial management among MSMEs indicates that digital literacy, financial knowledge, security awareness, and technology adoption can influence effective use of digital financial systems [7]. Therefore, simplicity and usability remain important considerations for the proposed application.

I. Future Enhancement

The proposed application provides a foundation for several future enhancements.

Automated Invoice Generation: The application can generate professional invoices directly from sales transactions. This can reduce repetitive data entry and provide users with structured transaction documents.

Digital Payment Integration: Integration with digital payment systems can simplify the recording of payments and connect payment information with business records. Previous research has explored accounting-based digital payment systems for SMEs to connect sales and payment information with financial reporting [6].

Cloud Synchronization: Cloud storage can allow business owners to access their ledger from multiple devices while maintaining a centralized data store.

Mobile and PWA Support: A Progressive Web Application can improve accessibility by allowing users to access the ledger through different devices. PWA-based approaches have previously been explored for SME data collection and business management [2].

AI-Based Financial Insights: Future versions can use machine learning or data-mining techniques to identify spending patterns, estimate future expenses, detect unusual transactions, and provide basic financial suggestions. AI and data-mining approaches have also been investigated for financial cost management in MSMEs [3].

Multilingual Support: The interface can be extended to support multiple regional languages, making the system more accessible to users with different language preferences.

GST and Tax Support: Future versions can include GST-related reports, tax summaries, and structured records that assist users in preparing information required for tax-related activities.

Business Analytics: Advanced dashboards can provide monthly comparisons, expense breakdowns, sales trends, customer dues, and other business indicators.

These enhancements can gradually transform the application from a basic digital ledger into a broader financial management platform for micro-enterprises.

V. CONCLUSIONS

The Micro Entrepreneur Ledger App provides a structured digital approach for managing the routine financial activities

of micro-entrepreneurs and small business owners. It brings business profiles, transactions, customer credit, payments, transaction history, and financial summaries together within a single application.

By keeping these activities within one system, the application helps users maintain their records in an organized manner and obtain a clearer view of their business information. The current implementation also provides a foundation for future additions such as cloud synchronization, digital payments, automated reporting, and financial analysis.

TABLE I
FUNCTIONAL EVALUATION OF THE MICRO ENTREPRENEUR LEDGER APP

Function	Evaluation Metric	Result
Transaction Recording	Entry Validation	successful
Sales and Expense Management	Calculation Accuracy	successful
Credit Management	Balance Accuracy	successful
Payment Tracking	Record Accuracy	successful
Transaction History	Retrieval Success	successful
Dashboard	Calculation Accuracy	successful

APPENDIX

The appendix provides additional details regarding the functional evaluation of the Micro Entrepreneur Ledger App.

The evaluation focuses on transaction recording, customer credit management, payment tracking, transaction history, balance calculation, and dashboard generation.

The final values in the evaluation table should be obtained through actual testing of the implemented application. The evaluation can be extended using additional indicators such as transaction processing time, successful validation rate, calculation accuracy, record retrieval success, and user satisfaction.

The test cases can be created using different business scenarios. For example, a test scenario may contain several sales transactions, multiple expenses, one purchase transaction, customer credit, and a subsequent customer payment.

The expected results can then be compared with the values displayed by the application. This process helps verify the correctness of the financial calculations and the consistency of stored records.

The evaluation can subsequently be performed across different transaction categories and business scenarios to understand the behavior of the application under different usage conditions.

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